

Multipath Interference Mitigation Based on Low-Complexity Adaptive High-Pass Filtering in Data Centre IM/DD Systems with OCS

H. N. Parajuli^{1,2}, Tonghui Ji^{1,2}, Xinping Wang^{1,2,3}, Ning Deng^{1,2*}, Shikui Shen⁴ and Jing Xu⁵

¹ School of Computing and Information Technology, Great Bay University, Guangdong, China

² Dongguan Key Laboratory for Intelligence and Information Technology, Guangdong, China

³ Department of EEE, Southern University of Science and Technology, Guangdong, China

⁴ China Unicom Research Institute, Building. 3, Haidian District, Beijing, China

⁵ Hainan Institute of Zhejiang University, Hainan, China

*Corresponding author: ndeng@gbu.edu.cn

Abstract We propose a low-complexity MPI mitigation scheme using adaptive high-pass filtering for data centre IM/DD systems. Experimental results demonstrate 4 dB improvement in SIR tolerance and 3.5 dB receiver sensitivity gain at HD-FEC threshold for a 56.88 Gbps PAM-4 signal in an OCS-integrated IM/DD link. ©2026 The Author(s).

Introduction

Intensity modulation with direct detection (IM/DD) remains a widely adopted technology in data centre (DCs) and artificial intelligence (AI) computing cluster (CCs) links due to its low cost and reduced power consumption. Multipath interference (MPI) is one of the major impairments in high-speed IM/DD system especially with multi-level pulse amplitude modulation (such as PAM-4) [1]. It is well known that MPI causes signal fluctuations and degrades system performance [2, 3]; thus, effective MPI mitigation is essential for high-speed PAM-4 IM/DD links.

In past years, numerous MPI mitigation techniques have been demonstrated [4-9]. Among them, finite impulse response (FIR)-based filtering is appealing due to its lack of training overhead and ease of implementation. However, conventional FIR-based high-pass filtering (FIR-HPF) [4-7] and moving-average filtering (referred to as the A1 algorithm in [4]) techniques require hundreds of taps or long window lengths, respectively, resulting in high computational complexity. In addition, the FIR-HPF necessitates careful tuning of the cut-off frequency and filter order, while the A1 requires proper adjustment of the window length to effectively suppress MPI.

In this paper, we propose a novel MPI mitigation technique based on adaptive high-pass filtering (A-HPF), offering ultra-low computational complexity per symbol, which requires only three multiplications and five additions. The filter cut-off frequency is adaptively adjusted according to signal fluctuations, enabling improved fluctuation tracking and suppression, which leads to enhanced performance. Moreover, as optical circuit switch (OCS) has been increasingly exploited in DC and CC optical systems [10], we investigate

the performance and effectiveness of our proposed A-HPF scheme in an IM/DD system with OCS, showing over 4 dB improvement in signal to interference ratio (SIR) tolerance at the HD-FEC BER threshold (3.8×10^{-3}).

Principle of Proposed Mitigation Scheme

Fig. 1(a) shows an OCS-integrated basic IM/DD link in DCs or CCs, where server transceivers are interconnected via optical fibers and an OCS. Imperfect connecting interfaces generate delayed replicas, which, upon photodetection with the main signal, cause fluctuation in the received signal, as illustrated in Fig. 1(a) insets (i) and (ii). To suppress MPI-induced fluctuation while preserving the data-bearing signal, we propose a first-order A-HPF scheme as shown in Fig. 1(b), described by the following equation

$$y[n] = x[n] - x[n-1] + \beta[n]y[n-1], \quad (1)$$

where $x[n]$ denotes the directly detected signal, $y[n]$ represents the filtered output, $\beta[n]$ is the adaptive pole coefficient satisfying $0 < \beta[n] < 1$ to ensure frequency response stability, and n is discrete time index. The first two terms in Eq. (1) correspond to the discrete-time differentiation of the input signal, which suppresses low frequency fluctuation while the third term ensures stability and filter shaping [11]. Further, by recursively substituting the delayed output term m times, for $m \rightarrow \infty$, Eq. (1) can be written as

$$y[n] \approx \sum_{k=0}^m (\prod_{j=0}^{k-1} \beta[n-j]) \{x[n-k] - x[n-k-1]\}, \quad (2)$$

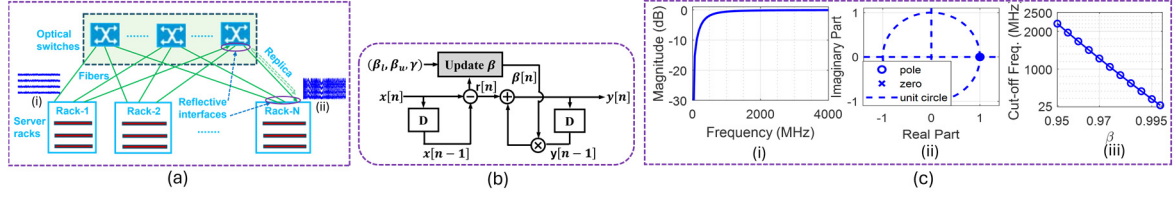


Fig. 1: a) Schematic of general OCS-based data centre IM/DD link showing the formation of delayed replica due to the reflective interfaces. b) Proposed adaptive HPF with recursive structure, and c) frequency response (i) of A-HPF for $\beta=0.992$ and corresponding pole-zero plot (ii). Mapping of β values to corresponding -3 dB cut-off frequencies (iii).

where, $k = 0, 1, \dots, m$ indexes the past input differences, and $j = 0, 1, \dots, k-1$ indexes the cumulative weight contributed by the time-varying coefficients $\beta[n-j]$. Whereas, in the fixed-coefficient HPF (F-HPF) case ($\beta[n] = \beta$), each input difference is weighted by β^{k+1} . Therefore, Eq. (1) inherently represents the output as a recursive accumulation of the input variation which enables significant reduction of arithmetic computation compared to conventional FIR-HPF and A1 methods.

The adaptive behavior of the filter is driven by the instantaneous fluctuation of the received signal. Considering the input difference defined as

$$r[n] = x[n] - x[n-1], \quad (3)$$

which measures the short-term fluctuation between successive samples. The adaptive pole coefficient is then updated as

$$\beta[n] = \beta_l + (\beta_u - \beta_l) \frac{\gamma}{|r[n]| + \gamma}, \quad (4)$$

where β_l and β_u define the allowable lower and upper bounds of the pole coefficient, respectively, and γ controls the adaptation sensitivity. When $|r[n]|$ is small, $\beta[n]$ approaches β_u , resulting in stronger suppression of slowly varying MPI-induced fluctuations and vice-versa. Consequently, the filter automatically adjusts its effective cut-off frequency according to the signal fluctuation characteristics.

Fig. 1(c)(i) shows the frequency response for $\beta = 0.992$ corresponding to -3 dB cut-off frequency of 330 MHz, confirming its ability to

suppress fluctuating components of low frequencies. Further, Fig. 1(c)(ii) depicts the pole-zero plot confirming the frequency response stability, and Fig. 1(c)(iii) illustrates the mapping of cut-off frequencies to β values. In Eq. (4), for a normalized signal $x[n] \in [-1, 1]$, $|r[n]| \in [0, 2]$. Then, assuming $\beta_u \approx 1$ (cut-off = 0 Hz) and $\gamma = E\{|r[n]|\} = 1$, only β_l should be specified. As an example, for a given signal, to track and suppress fluctuations in the range of 0-2200 MHz, $\beta_l = 0.95$ and $\beta_u = 1$ can be chosen as shown in Fig. 1(c)(iii). Moreover, as linear time-invariant (LTI) filters, FIR methods introduce group delay that requires compensation, whereas the proposed A-HPF, being time-varying, relaxes this constraint.

Experimental Setup

Fig. 2 illustrates the experimental setup. A 56.88 Gbps offline-generated PAM-4 waveform (PRBS $2^{18}-1$, RRC roll-off 0.1) was uploaded to an arbitrary waveform generator (AWG, Keysight M8194A), amplified, and applied to a 25 GHz Mach-Zehnder modulator (MZM, FTM7938EZ). A 1550 nm DFB laser (16 dBm, 2 MHz linewidth) served as the optical carrier. The modulated signal was launched into a piece of 1-km single-mode fibre (SMF-1) and then connected to a 32×32 OCS (Polatis), where port 1 was optically switched to port 5, and transmitted through another 1 km of SMF-2. The internal return loss of the OCS was measured as low as < -40 dB. However, relatively strong MPI may still occur due to the additional optical connectors which was emulated using an 80:20 coupler and a delayed branch of 2-km SMF-3. We adopted a variable optical attenuator (VOA-1) to control the SIR, and a PC-2 to study the worse performance of the interference. The recombined signal was directly

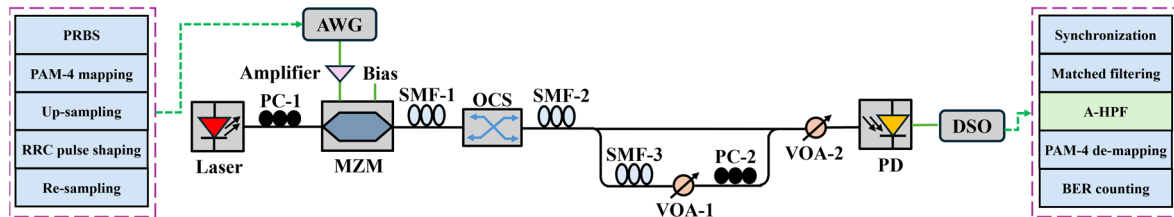


Fig. 2: Experimental setup for evaluating the A-HPF performance. AWG: arbitrary waveform generator, MZM: Mach-Zehnder modulator; OCS: optical circuit switch; SMF: single-mode fiber; PC: polarization controller; VOA: variable optical attenuator; PD: photodetector; DSO: digital sampling oscilloscope; A-HPF: adaptive recursive high-pass filter.

detected by a photodetector (PD, XPDV3x20R), and the waveform was captured by a digital sampling oscilloscope (DSO). The acquired data were processed offline followed by BER evaluation under varying SIR and received optical power (ROP) conditions. To capture the statistical properties of MPI, each measurement was repeated for five times, and the average results are reported.

Results and Discussion

The filter parameters of the considered filtering techniques were initially optimized by minimizing the BER for a signal received at an ROP of -2 dBm and an SIR of 22 dB. The optimal β for the F-HPF is found to be 0.992 (cut-off=330 MHz), as shown in Fig. 3(a). For the A-HPF case, as illustrated in Fig. 3(b), with $\beta_u = 1$ and $\gamma = 1$ the optimal value for β_l is 0.95. Similarly, the optimal FIR-HPF parameters were obtained as a cut-off frequency of 350 MHz and a tap length of 500, while the optimal window size for the A1 method was 250. These parameters are used in the subsequent analysis.

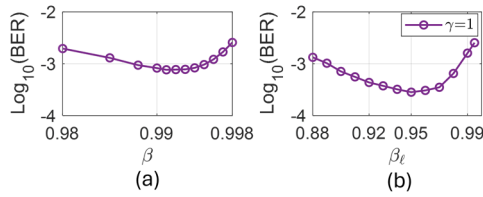


Fig. 3. Filter parameter optimization through BER minimization: (a) fixed β case (F-HPF); (b) adaptive HPF (A-HPF) showing optimization of β_l for fixed $\gamma = 1$.

Figs. 4(a) and (b) depict the measured BER as a function of ROP and SIR respectively for various schemes. Fig. 4 (a) demonstrates that the proposed A-HPF method achieves a ~ 3.5 dB receiver sensitivity gain at $\text{SIR} > 22$ dB, evaluated at the HD-FEC threshold. At a fixed ROP of -2 dBm, the BER reduces from 5.1×10^{-3} to 2.85×10^{-4} (corresponding to an approximately 18-fold reduction), thereby demonstrating effective mitigation of MPI-induced degradation. Likewise, as shown in Fig. 4(b), at HD-FEC, 4 dB improvement in SIR tolerance is achieved. For $\text{SIR} > 22$ dB, KP4-FEC threshold is achieved. For very low MPI power (< -28 dBm, corresponding to $\text{SIR} > 26$ dB), where MPI becomes negligible, the HPF suppresses low-frequency components of the signal, thereby inducing baseline wander, which results in waveform distortion and degraded performance [4, 9]. Figs. 4(c) and (d) show the 1 Sa/symbol signal before and after mitigation, respectively, using the proposed A-HPF for a received signal with a SIR of 22 dB, demonstrating the effectiveness of the proposed method in suppressing MPI-induced fluctuations. Further, Figs. 4(a) and (b) demonstrate that the proposed A-

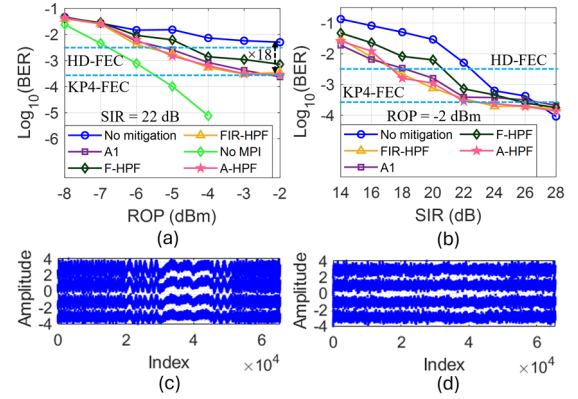


Fig. 4. Measured BER performance of various filtering schemes: (a) BER vs. ROP at SIR = 22 dB; (b) BER vs. SIR at ROP = -2 dBm; (c) 1 Sa/symbol samples before mitigation (SIR = 22 dB); and (d) after mitigation using A-HPF.

HPF achieves BER performance comparable to the finely-tuned A1 and FIR-HPF methods.

Moreover, Table 1 summarizes the computational complexity required by each method to achieve the results presented in Figs. 4(a) and (b), showing that the proposed scheme requires an ultra-low number of computations per symbol while effectively mitigating MPI.

Table 1: Comparison of the calculated complexity per symbol among the proposed and previously reported methods.

Methods	Parameter size	Addition/subtraction	Mul. / div.
FIR-HPF [4-7]	500	500	501
A1 [4]	250	249	1
A-HPF (this work)	-	5	3

Summary

We have proposed and experimentally validated an A-HPF scheme for data centre IM/DD systems that effectively tracks and suppresses MPI-induced fluctuations with low complexity. The A-HPF is realized using a simple first-order recursive structure with an adaptive pole-tuning mechanism. The developed pole-tuning technique exploits the instantaneous fluctuation characteristics of the received signal, enabling robust tracking and effective mitigation. Experimental results in a PAM-4 IM/DD system with OCS demonstrate a 4 dB improvement in SIR tolerance and a 3.5 dB receiver sensitivity gain at the HD-FEC threshold. Notably, the proposed scheme achieves this performance with a fixed computational complexity of only five additions and three multiplications, significantly reducing the number of computations compared to conventional FIR-HPF and A1 methods. These results demonstrate the strong suitability of the proposed scheme for mitigating MPI in practical data centre IM/DD systems.

Acknowledgments

This work was partially supported by the National Natural Science Foundation of China (NSFC) (W2533179, 62571085), the Guangdong Research Team for Communication and Sensing Integrated with Intelligent Computing (2024KCXTD047), and the Hainan Provincial Foreign Expert Projects (G20230607010E).

References

- [1] C. R. S. Fludger, M. Mazzini, T. Kupfer and M. Traverso, "Experimental measurements of the impact of multi-path interference on PAM signals," *OFC 2014*, San Francisco, CA, USA, 2014, pp. 1-3, doi: 10.1364/OFC.2014.W1F.6.
- [2] L. Fei, Y. Xu, Y. Guo, W. Hu and Q. Zhuge, "Nonuniform Error Distributions Induced by Multipath Interference in Optical IM/DD PAM4 Systems: Implications for FEC Performance," in *Journal of Lightwave Technology*, vol. 43, no. 24, pp. 10947-10959, 2025, doi:10.1109/JLT.2025.3618678.
- [3] H. N. Parajuli, X. Wang, N. Deng *et al.*, "Multipath Interference in High-speed IMDD Optical Links: Modelling and Mitigation Techniques," in *Chinese Optics Letters*, vol. 24, no. 7, 2026, doi:10.3788/COL202624.070005.
- [4] C. Huang, H. Song, L. Dai *et al.*, "Optical Multipath Interference Mitigation for High-Speed PAM4 IMDD Transmission System," in *Journal of Lightwave Technology*, vol. 40, no. 16, pp. 5490-5501, 2022, doi: 10.1109/JLT.2022.3181695.
- [5] W. Ni, Y. Liu, D. Zou *et al.*, "Low-Complexity Long-Term Memory Postfilter Assisted Mitigation of Optical Multipath Interference in PAM-4 IM-DD Systems," in *Journal of Lightwave Technology*, 2026, doi: 10.1109/JLT.2025.3645344.
- [6] K. Lian, Z. Xu, M. Qiao *et al.*, "Optical Multi-Path Interference Mitigation for PAM4-IMDD Systems Using Balanced Coding," in *Journal of Lightwave Technology*, vol. 42, no. 17, pp. 5918-5927, 2024, doi: 10.1109/JLT.2024.3403161.
- [7] Y. Qiu, M. Xiang, J. Xiang *et al.*, "Feedforward Neural Network Enabled Optical Multi-Path Interference Mitigation for High-Speed IM-DD Transmissions," in *Journal of Lightwave Technology*, vol. 42, no. 19, pp. 6749-6759, 2024, doi: 10.1109/JLT.2024.3412654.
- [8] Y. Li, Z. Wei, B. Deng *et al.*, "Adaptive decision threshold for an optical multipath-interference-impaired short-reach 50-Gbps PAM4 transmission," in *Optics Letters*, vol. 48, no. 21, pp. 5675-5678, 2023, doi:10.1364/OL.500244.
- [9] N. Cheng, D. Zhang, D. Ge *et al.*, "Optical Multipath Interference Mitigation for PAM4 Transmission Using Line Coding and High-pass Filtering," *2023 Optical Fiber Communications Conference and Exhibition (OFC)*, San Diego, CA, USA, 2023, pp. 1-3, doi: 10.1364/OFC.2023.W2A.24.
- [10] N. Deng, Z. Chang, and R. Chen, "Optical Switching for Inter and Intra Computing Center Networks," paper Mo3K-1, CLEO-PR, 2024, doi: 10.1109/CLEO-PR60912.2024.10676873.
- [11] A. V. Oppenheim and R. W. Schaffer, "Structures for discrete time systems," in *Discrete-Time Signal Processing*, 3rd ed., Pearson, 2010, ch. 6.